3.2 short run costs

DEACC
CACEC
BCECA
DEBEC
CABDB
CCBDB
A

The following question is based on the output and cost data in the table below.

Quantity of Output Produced (units)	Total Variable Cost ($)	Total Cost ($)
1	100	200
2	190	290
3	270	370
4	340	440
5	420	520
6	510	610
7	610	710

1. According to the cost schedule in the table, the firm's total fixed cost is

(A) increasing
(B) decreasing
(C) $90
(D) $100 ✓
(E) impossible to determine

2. A competitive firm produces a product using labor and plastic. The firm is initially in equilibrium. If the cost of plastic suddenly increases, which of the following will occur?

(A) The demand curve for the product will shift to the left.
(B) The firm's demand curve for plastic will shift to the left.
(C) The firm will increase the number of units offered for sale.
(D) The firm will definitely go out of business, since competitive firms earn zero economic profits in equilibrium.
(E) The firm's marginal costs will increase at each level of output. ✓

3. Which of the following statements about short-run costs is always true as a firm increases its production?

(A) Average fixed cost decreases. ✓
(B) Average variable cost decreases.
(C) Average total cost decreases.
(D) Marginal cost is lower than the average variable cost.
(E) Marginal cost is lower than the average total cost.

3.2 short run costs

Answer A

Correct. Since average fixed cost is calculated as total fixed cost/output and total fixed cost does not change as output changes, an increase in output must decrease average fixed cost.

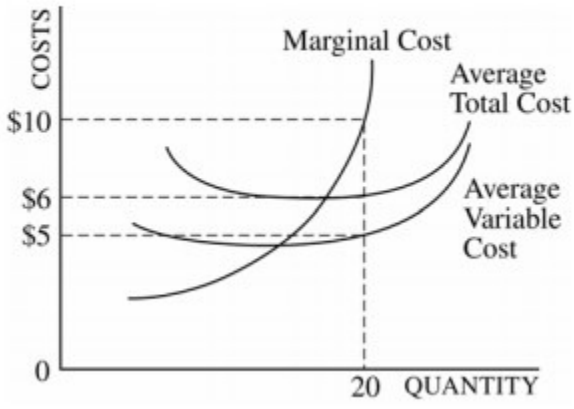
4. As output of a firm increases, the difference between the firm's average total cost and its average variable cost gets smaller because the firm's

(A) total cost is increasing
(B) marginal cost is increasing
(C) average fixed cost is decreasing ✓
(D) marginal product of labor is decreasing
(E) long-run average total cost is decreasing

5. A firm is producing 100 units of output at a total cost of $400. The firm's average variable cost is $3 per unit. What is the firm's total fixed cost?

(A) $1
(B) $50
(C) $100 ✓
(D) $300
(E) $400

6.



The graph above shows the cost curves for a competitive firm that produces 20 units of output. What are the total cost and the total fixed cost of producing 20 units of output?

3.2 short run costs

(A)

Total Cost	Total Fixed Cost
$10	$0

(B)

Total Cost	Total Fixed Cost
$120	100

(C) ✓

Total Cost	Total Fixed Cost
$120	$20

(D)

Total Cost	Total Fixed Cost
$200	$100

(E)

Total Cost	Total Fixed Cost
$200	$20

7. An increase in which of the following will cause a firm's marginal cost curve to shift upward?

(A) The price of a variable input ✓
(B) The price of a fixed input
(C) The level of output
(D) Labor productivity
(E) The demand for the firm's product

8. Assume that a firm uses only one variable input. If a firm is experiencing diminishing returns, which of the following is true as more of the variable input is used?

3.2 short run costs

(A) Marginal cost will decrease at a constant rate.
(B) Marginal cost will decrease at a diminishing rate.
(C) Marginal cost will increase. ✓
(D) Marginal product will increase at a constant rate.
(E) Marginal product will increase at a diminishing rate.

9. Assume that a profit-maximizing firm is perfectly competitive in both the output and the factor markets and is at its long-run equilibrium. The firm's output is 100 units, its total revenue is $600.00, and the fixed cost of production is $50.00. Based on this information, which of the following is true for the firm?

(A) Its marginal cost is $5.50, and its average total cost is $5.50.
(B) Its marginal cost is $5.50, and its average variable cost is $5.50.
(C) Its marginal cost is $6.00, and its average total cost is $5.50.
(D) Its marginal cost is $6.00, and its average fixed cost is $5.50.
(E) Its marginal cost is $6.00, and its average variable cost is $5.50. ✓

10. Assume that total fixed costs are $46, that the average product of labor is 5 units when 10 units of output are produced, and that the wage rate is $12. If labor is the only variable input, what is the average total cost of producing 10 units of output?

(A) $2
(B) $5
(C) $7 ✓
(D) $9
(E) $12

11. At 100 units of a firm’s output, average total cost is $10, average variable cost is $8, average fixed cost is $2, and marginal cost is $12. How will each of the following change as the firm's output further increases?

3.2 short run costs

(A)

Average Total Cost	Average Variable Cost	Average Fixed Cost
Increase	Increase	Increase

(B) ✓

Average Total Cost	**Average Variable Cost**	**Average Fixed Cost**
Increase	Increase	Decrease

(C)

Average Total Cost	**Average Variable Cost**	**Average Fixed Cost**
Increase	Decrease	Decrease

(D)

Average Total Cost	**Average Variable Cost**	**Average Fixed Cost**
Decrease	Increase	Increase

(E)

Average Total Cost	**Average Variable Cost**	**Average Fixed Cost**
Decrease	Decrease	Decrease

12. At 100 units of output, a firm's total cost is $10,000. If the firm's total fixed cost is $4,000, its average variable cost is equal to

(A) $140
(B) $100
(C) $60 ✓
(D) $40
(E) $0

3.2 short run costs

13.

Quantity	Marginal Cost	Fixed Cost
0	—	20
1	15	—
2	10	—
3	15	—

According to the table above, which shows the costs of production for a firm, the average total cost of producing 3 units of output is

(A) 5.00
(B) 11.67
(C) 13.33
(D) 15.00
(E) 20.00 ✓

14. In the short run, which of the following is true of a firm's average total cost of production?

(A) It is equal to marginal cost plus average variable cost.
(B) It is equal to marginal cost plus average fixed cost.
(C) It is equal to average fixed cost plus average variable cost. ✓
(D) It always increases when a firm increases production.
(E) It is zero if the firm shuts down.

15. Which of the following must be true if at the tenth unit of output, marginal cost (MC) is $130 and average total cost (ATC) is $150 ?

(A) ATC of producing the ninth unit is higher than $150. ✓
(B) ATC of producing the ninth unit is less than $150.
(C) MC of producing the ninth unit is higher than $130.
(D) The average variable cost of producing the tenth unit is higher than $150.
(E) The average variable cost of producing the tenth unit is equal to $20.

Answer A

Correct. When ATC is greater than MC, the marginal cost is rising and the ATC is falling with increases in output. Producing one less unit of output raises ATC; therefore, the ATC of producing the ninth unit is greater than $150.

3.2 short run costs

16. Which of the following is always true of the relationship between average and marginal costs?

(A) Average total costs are increasing when marginal costs are increasing.

(B) Marginal costs are increasing when average variable costs are higher than marginal costs.

(C) Average variable costs are increasing when marginal costs are increasing.

(D) Average variable costs are increasing when marginal costs are higher than average variable costs. ✓

(E) Average total costs are constant when marginal costs are constant.

17. Which of the following is true about a firm's average variable cost?

(A) It will rise if marginal cost is less than average variable cost.

(B) It will never equal the firm's marginal cost.

(C) It will decline when the firm's marginal product declines.

(D) It will be negative if marginal revenue declines.

(E) It will equal average total cost when fixed costs are zero. ✓

18. Beyond a certain level of output, the short-run marginal cost will rise because

(A) there is no fixed input and costs will increase

(B) at least one input is fixed and eventually diminishing returns will occur ✓

(C) the cost of the variable input increases when marginal product increases

(D) the demand for the good decreases when production is limited

(E) input prices increase when production increases and consumption is limited

---

The following questions are based on the table below, which shows a firm's average variable cost and average total cost.

Output	Average Variable Cost	Average Total Cost
1	$40	$160
2	35	95
3	40	80
4	45	75
5	50	74
6	55	75

3.2 short run costs

19. The average fixed cost of producing four units of output is equal to

(A) $120
(B) $95
(C) $75
(D) $50
(E) $30 ✓

20.

Units of Labor Input	Units of Output
1	8
2	20
3	30

The table above shows the amount of labor inputs necessary to produce given levels of output. If the cost of a unit of labor is $20 and total fixed cost is $100, the average total cost of producing 20 units of output is

(A) $1
(B) $2
(C) $7 ✓
(D) $40
(E) $120

The following questions are based on the table below, which gives cost information for a perfectly competitive firm.

Quantity	Average Fixed Costs	Average Variable Costs	Marginal Costs
0			
1	$100.00	$55.00	$55.00
2	50.00	45.00	35.00
3	33.33	50.00	60.00
4	25.00	55.00	70.00
5	20.00	60.00	80.00
6	16.67	65.00	90.00

21. The average total cost to the firm of producing 2 units of output is

3.2 short run costs

(A) $35.00
(B) $85.00
(C) $95.00 ✓
(D) $100.00
(E) $130.00

Units of Labor	Quantity of Output	Total Cost
0	0	$100
1	10	$120
2	25	$150

22. Given the information, the average variable cost of 25 units of output is

(A) $2 ✓
(B) $6
(C) $25
(D) $50
(E) $75

The table below shows a competitive firm's total variable cost (TVC) and total fixed cost (TFC) at various units of output.

Output	TVC	TFC
0	0	9
1	3	9
2	8	9
3	15	9
4	26	9
5	38	9

23. When output is 3 units, which of the following is correct?

3.2 short run costs

(A)

Average Variable Cost	Marginal Cost
5	6

(B) ✓

Average Variable Cost	Marginal Cost
5	7

(C)

Average Variable Cost	Marginal Cost
6	5

(D)

Average Variable Cost	Marginal Cost
7	7

(E)

Average Variable Cost	Marginal Cost
7	6

---

24. If the marginal cost of producing the first unit of some good is $20 and the marginal cost of producing the second unit is $30, the average variable cost of producing 2 units is

(A) $5
(B) $10
(C) $20
(D) $25 ✓
(E) $50

3.2 short run costs

25.

Quantity (units)	Total Cost
0	$30
1	40
2	47
3	51
4	59

The table above shows a firm's total cost of producing various units of output. What is the average variable cost of producing three units?

(A) $4

(B) $7 ✓

(C) $10

(D) $17

(E) Cannot be determined from the given information

26.

Quantity of Output (units)	Total Variable Cost
0	$0
1	$40
2	$50
3	$65
4	$90

Assume that the fixed cost is $50. Based on the cost and output data in the table above, what is the marginal cost when the firm increases its output from three to four units and the average total cost of producing 4 units?

3.2 short run costs

(A)

Marginal Cost	Average Total Cost
$35	$40

(B)

Marginal Cost	Average Total Cost
$35	$35

(C) ✓

Marginal Cost	Average Total Cost
$25	$35

(D)

Marginal Cost	Average Total Cost
$25	$25

(E)

Marginal Cost	Average Total Cost
$10	$25

27. If labor is the only variable input and it costs $15 per hour and if the marginal product of labor is 3 units per hour, the short-run marginal cost of 1 unit of output is approximately

(A) $0.20
(B) $3.00
(C) $5.00 ✓
(D) $15.00
(E) $45.00

3.2 short run costs

The following questions are based on the table provided.

The table provided shows a firm's average fixed cost (AFC), total variable cost (TVC), and total economic profit at different levels of output in units.

Output	AFC	TVC	Profit
50	$20		
200		$1,600	$1,000

28. When the firm produces 200 units, the AFC is

(A) $\$0$

(B) $\$5$ ✓

(C) $\$8$

(D) $\$20$

(E) $\$80$

Answer B

Correct. Since the average fixed costs $(\text{AFC})=\$20$ when output equals 50 units, total fixed costs can be calculated as $\text{AFC} \times \text{Q} = \$20 \times 50 = \$1{,}000$. Thus, when output equals 200 units, AFC is $\$1{,}000/200=\$5$.

29. Over the entire range of output for which a firm's short-run average variable cost is decreasing, which of the following must be true?

(A) Short-run marginal cost is falling.

(B) There are increasing returns to scale.

(C) There are decreasing returns to scale.

(D) Short-run marginal cost is less than short-run average variable cost. ✓

(E) Short-run marginal cost is increasing due to diminishing returns.

3.2 short run costs

Answer D

Correct. Short-run marginal cost represents the change in total variable cost for a one unit change in output. When short-run marginal cost is less than short-run average variable costs at a given level of output, the short-run average variable cost must be falling.

30. Short-run marginal costs eventually increase because of the effects of

(A) increasing marginal product

(B) diminishing marginal product ✓

(C) diseconomies of scale

(D) economies of scale

(E) increasing fixed costs

31. If a firm's production function exhibits diminishing marginal product of the variable input in the short run, which of the following about the firm's short-run marginal cost (MC) curve must be true?

(A) As output increases, the MC curve slopes upward. ✓

(B) As output increases, the MC curve slopes downward and becomes flatter.

(C) As output increases, the MC curve slopes downward and becomes steeper.

(D) The MC curve is horizontal.

(E) The MC curve is vertical.

32. For a firm where labor is the only variable input, which of the following happens when diminishing returns set in?

(A) Average variable cost begins to increase.

(B) Average product of labor begins to decline.

(C) Total product begins to decline.

(D) Marginal cost begins to increase. ✓

(E) Average total cost begins to increase.

33. In the short run, assume diminishing marginal product of labor sets in with the hiring of the second worker. Which of the following will remain constant as a firm produces more output?

(A) Average product of labor

(B) Total cost

(C) Total fixed cost ✓

(D) Total variable cost

(E) Marginal cost

3.2 short run costs

Answer C

Correct. Total fixed cost will remain unchanged as output increases (or decreases) in the short run, since the firm cannot add more capital resources in the short run.

34. A graph shows quantity of output produced on the horizontal axis and shows total variable cost and total cost on the vertical axis. Which of the following is true about the vertical distance between total variable cost and total cost as output increases?

(A) It decreases.

(B) It remains constant. ✓

(C) It increases.

(D) It decreases and then increases.

(E) It increases and then decreases.

Answer B

Correct. The vertical distance between total variable cost and total cost at each output level is total fixed cost, which is constant.

35. FormalWear is a clothing factory that hires workers as a variable input and rents sewing machines as a fixed input. Which of the following will happen when the marginal product of workers decreases?

(A) Total variable cost will decrease.

(B) Total fixed cost will increase.

(C) Short-run average total cost will decrease.

(D) Marginal cost will increase. ✓

(E) Marginal cost will equal average total cost.

Answer D

Correct. The marginal cost function is the “mirror image” of the marginal product of labor function; therefore, when the marginal product of workers (labor) decreases, marginal cost must increase.

3.2 short run costs

36.

Number of Cases	Total Cost ($)
0	15
1	25
2	45
3	75
4	115
5	165

The table provided shows the daily total cost data for Hansel's Hot Sauce. If the average fixed cost was $5 per case, how many cases of hot sauce had Hansel's Hot Sauce produced?

(A) 1

(B) 2

(C) 3 ✓

(D) 4

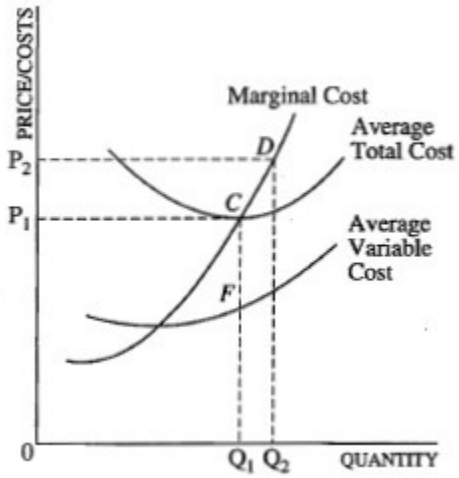
(E) 5

Answer C

Correct. Since total cost is $15 when 0 cases are produced, total fixed cost is $15. For average fixed cost to be $5 per case, 3 cases must be produced (since average fixed cost is $\frac{\$15}{3} = \$5$ only when 3 cases are produced).

3.2 short run costs

The following questions refer to the graph below, which shows the cost curves for a profit maximizing, perfectly competitive firm.



37. The vertical distance *CF* represents the

(A) total cost of producing Q_1 units of output
(B) average total cost of producing Q_1 units of output
(C) average fixed cost of producing Q_1 units of output ✓
(D) average variable cost of producing Q_1 units of output
(E) amount of the firm's loss resulting from producing Q_1 units of output

38. If a firm's average total cost decreases as the firm increases its output, the firm's marginal cost must be

(A) greater than the average variable cost
(B) less than the average fixed cost
(C) less than the average total cost ✓
(D) decreasing
(E) negative

39. If a new tax on capital increases a firm's fixed cost of production, which of the following will occur in the short run?

3.2 short run costs

(A) Marginal cost will increase.
(B) Average variable cost will increase.
(C) Average total cost will increase. ✓
(D) The profit-maximizing level of output will increase.
(E) The profit-maximizing level of output will decrease.

40. If the average variable cost of producing 5 units of a good is $100 and the average variable cost of producing 6 units is $150, then the marginal cost of increasing output from 5 to 6 units is

(A) $50
(B) $250
(C) $300
(D) $400 ✓
(E) $500

41. If the price of a firm's variable input increases, which of the following will occur?

(A) The firm will decrease its level of production. ✓
(B) The price of the good will decrease in the short run.
(C) The firm's marginal costs will decrease at every level of output.
(D) The firm's average fixed cost will decrease.
(E) More firms will enter the industry in the long run.

42. Which of the following factors can cause a firm's cost curves to shift upward?

(A) An increase in wages ✓
(B) An increase in the firm's output
(C) An increase in the output price
(D) A decrease in the firm's output
(E) A decrease in the price of energy

43. In the short run, a profit-maximizing firm, faced with U-shaped average cost curves, is producing a level of output at which the average total cost of production is minimized. At this level of output, which of the following is true for the firm?

(A) Marginal cost equals average fixed cost.
(B) Marginal cost equals average variable cost.
(C) Marginal cost equals average total cost. ✓
(D) Profit per unit equals average total cost.
(E) Profit per unit equals marginal cost.

3.2 short run costs

44. In the short run, which of the following costs must continuously decrease as output produced increases?

(A) Total variable cost

(B) Total fixed cost

(C) Average variable cost

(D) Average fixed cost ✓

(E) Average total cost

The table below is partially filled in with the different types of costs for a firm.

Quality	Variable Cost	Fixed Cost	Total Cost	Average Total Cost
0	$0	$60		
1			$130	
2				$90

45. Based on the information in the table, what is the marginal cost of producing the second unit?

(A) $50 ✓

(B) $60

(C) $70

(D) $90

(E) $180

Answer A

Correct. The variable cost for the first unit is $\$130 - \$60 = \$70$. The variable cost for the second unit is $\$180 - \$60 = \$120$. Therefore, the marginal cost of producing the second unit is $(\$120 - \$70)/1 = \$50$.

---

46. Assume that the short-run marginal cost curve initially falls, and it then rises as quantity of output increases. Which of the following must be true?

3.2 short run costs

(A) The marginal product of labor is always falling.
(B) At every output level, the short-run marginal cost curve is always above the average variable cost curve.
(C) As output increases, the unit price of labor (the wage rate) is first falling and then rising.
(D) Initially the marginal product of labor increases but eventually marginal product of labor decreases. ✓
(E) The average variable cost curve must always be rising.

Answer D

Correct. This is the typical U-shaped short-run marginal cost curve. Short-run marginal cost (MC) depends on the wage rate and the marginal product of labor (MP_L). The relationship is $\text{MC} = \frac{\text{Wage}}{\text{MP}_\text{L}}$. If the MP_L is increasing, then MC will decrease, and vice versa. With the typical U-shaped MC curve, the MP_L increases initially, which causes MC to decline initially. But eventually diminishing returns set in, which causes MC to eventually increase.

47. Marginal cost is defined as the

(A) change in total cost resulting from producing an additional unit of output ✓
(B) change in total cost resulting from using an additional unit of input
(C) difference between total cost and total variable cost
(D) difference between total variable cost and total fixed cost
(E) difference between average total cost and average variable cost divided by output

The chart below gives a firm's total cost of producing different levels of output.

Output	Total Cost
0	$13
1	20
2	25
3	28
4	32
5	43
6	60

48. The marginal cost of producing the fourth unit of output is

3.2 short run costs

(A) $ 4 ✓

(B) $11

(C) $19

(D) $32

(E) impossible to determine from the information given

49. The total variable cost of producing five units of output is

(A) $ 6

(B) $11

(C) $30 ✓

(D) $43

(E) impossible to determine from the information given

---

50. When the marginal cost curve lies below the average total cost curve, it is true that as output increases

(A) marginal cost is decreasing

(B) marginal cost is increasing

(C) average total cost is decreasing ✓

(D) average total cost is increasing

(E) average variable cost is decreasing

51. Which of the following is true of the marginal cost curve?

(A) It intersects the average variable cost curve and the average total cost curve at each curve's minimum point. ✓

(B) It intersects the average variable cost curve and the average fixed cost curve at each curve's minimum point.

(C) It lies between the total cost curve and the total variable cost curve.

(D) It increases initially, for a time, but begins to decline when the point of diminishing returns is reached.

(E) It decreases, because average variable cost is less than marginal cost.

52. Which of the following best explains why a firm's short-run marginal cost curve shifts down when it purchases new, more efficient equipment and experiences an increase in its total cost?

3.2 short run costs

(A) The situation represents an exception to the law of diminishing returns.

(B) The average total cost curve shifts upward as a result of the equipment purchase, and there is a movement up along the marginal cost curve.

(C) The equipment purchase is a fixed cost, and the new equipment will cause a reduction in the cost of producing each additional unit. ✓

(D) The average variable cost curve shifts upward as a result of the equipment purchase, and the marginal cost curve shifts downward.

(E) The marginal cost curve shifts downward because of the law of diminishing returns.

53. Which of the following statements about short-run costs is true?

(A) Average fixed cost plus variable cost equals total cost.

(B) Average total cost plus average fixed cost equals average variable cost.

(C) Total fixed cost increases in constant increments as output produced increases.

(D) Total fixed cost plus total variable cost equals total cost. ✓

(E) At low output levels, as output increases, total fixed cost and total variable cost decrease, reducing average total cost.

Answer D

Correct. In the short run, a firm has both fixed and variable costs, the sum of which is equal to total cost. Also, average fixed cost plus average variable cost will equal average total cost.

54. At a firm's current output level, average fixed cost is $10, average variable cost is $30, average total cost is $40, and marginal cost is $55. Which of the following must be true?

(A) Average fixed cost, average variable cost, and average total cost are all decreasing.

(B) Average fixed cost, average variable cost, and average total cost are all increasing.

(C) Average fixed cost and average total cost are both decreasing, while average variable cost is increasing.

(D) Average fixed cost is decreasing, and both average variable cost and average total cost are increasing. ✓

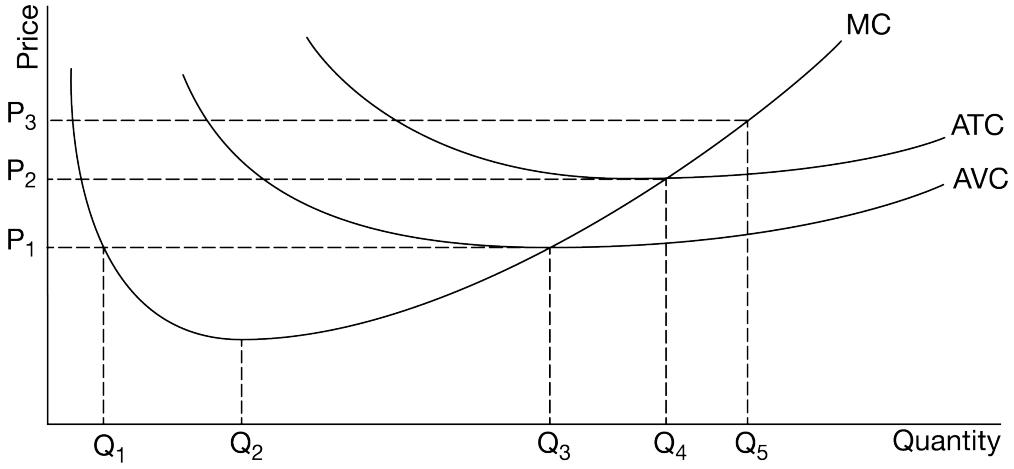
(E) Average fixed cost and average variable cost are both decreasing, while average total cost is increasing.

Answer D

Correct. Marginal cost is above average variable cost and average total cost, causing both to increase. Average fixed cost always decreases.

3.2 short run costs

55.



The graph above shows per unit cost information for firm X. At what quantity of output do diminishing returns begin for firm X?

(A) Q_1

(B) Q_2 ✓

(C) Q_3

(D) Q_4

(E) Q_5

Answer B

Correct. Diminishing returns begin at the point where marginal cost is minimized because the next unit of output will be produced at a higher cost, which results from diminishing returns.

3.2 short run costs

56.

Number of Workers	Number of Units of Capital	Total Product (Quantity)
0	5	0
1	5	1
2	5	3
3	5	7
4	5	10
5	5	12

Suppose labor and capital can be hired in a competitive market. The wage is $\$15$ per worker and the rental rate of capital is $\$10$ per unit. Based on the table, what are the average variable cost (AVC) and average total cost (ATC) of producing 10 units of output?

(A) AVC is $\$5$ and ATC is $\$12$

(B) AVC is $\$6$ and ATC is $\$11$ ✓

(C) AVC is $\$15$ and ATC is $\$25$

(D) AVC is $\$60$ and ATC is $\$110$

(E) AVC is $\$105$ and ATC is $\$175$

Answer B

Correct. When the firm produces 10 units, it hires 4 workers at a wage rate of $\$15$. Therefore, its variable cost is $\$60$ $(=4\times\$15)$, and the average variable cost is $\$6\left(=\frac{\$60}{10}\right)$. Total cost is the sum of variable cost and fixed cost. The firm rents capital (the fixed inputs) for $\$10$ per unit. Therefore, fixed cost is $\$50$ $(=5\times\$10)$, total cost is $\$110$ $(=\$60+\$50)$, and average total cost is $\$11\left(=\frac{\$110}{10}\right)$.

57. Which of the following best describes the relationship between the average total cost curve and the marginal cost curve in the short run?

(A) If the average total cost curve is rising, the marginal cost curve is above the average total cost curve. ✓

(B) If the average total cost curve is rising, the marginal cost curve is below the average total cost curve.

(C) If the average total cost curve is above the marginal cost curve, the marginal cost curve is rising.

(D) If the average total cost curve is below the marginal cost curve, the marginal cost curve is falling.

(E) If the average and marginal cost curves intersect, the marginal cost curve is at a minimum.

58. Technological advances will lead to

3.2 short run costs

(A) an increase in marginal utility

(B) a decrease in average total costs ✓

(C) a decrease in net exports

(D) an increase in marginal costs

(E) diseconomies of scale

59. Which of the following best explains why the short-run average total cost curve is U-shaped?

(A) Spreading total fixed costs over a larger output, and constant returns

(B) Spreading total fixed costs over a larger output, and eventually diminishing returns ✓

(C) Increasing total fixed costs and increasing returns

(D) Increasing average variable costs and decreasing returns

(E) Decreasing average variable costs and increasing returns